Name – Mukund Raj

Roll No. - 23b4245

Summer of Code 2026 – Week 2 Assignment

Introduction to Neural Networks and Medical Imaging

Neural Networks

A neural network is a machine learning model inspired by the human brain. It consists of connected layers of artificial neurons that learn patterns from data. Instead of following fixed rules, neural networks improve their performance by learning from examples.

Neural networks are widely used in image recognition, speech processing, language translation, and many other AI applications.

How Neural Networks Learn

Neural networks learn by making predictions and comparing them with the correct answers. If the prediction is wrong, the network adjusts its internal weights to reduce the error. By repeating this process many times, the model gradually becomes more accurate.

Backpropagation

Backpropagation is the learning process used by neural networks. It calculates how much each neuron contributed to the error and updates the network accordingly. This allows the model to improve its predictions over time.

Convolutional Neural Networks (CNNs)

CNNs are a special type of neural network designed for image data. They automatically learn important visual features such as edges, textures, shapes, and patterns from images.

CNNs are widely used for tasks such as image classification, object detection, and image segmentation.

CNNs in Healthcare

CNNs have become very important in medical imaging. They can analyse X-rays, MRI scans, CT scans, and other medical images to help detect diseases and abnormalities. These systems can assist doctors by improving diagnostic accuracy and reducing the time required to analyse large numbers of medical images.

Image Processing in Python

For this assignment, I used a chest X-ray image and a brain MRI image to perform several basic image processing operations.

1. Grayscale Conversion

The images were converted from color to grayscale. This removes color information and keeps only intensity values, making image processing simpler and more efficient.

2. Histogram Visualization

Histograms were generated for both images to observe the distribution of pixel intensities. Histograms help in understanding the brightness and contrast characteristics of an image.

3. Image Resizing

The images were resized to a fixed resolution. Resizing is commonly used before training machine learning models because neural networks usually require images of a consistent size.

4. Edge Detection

Edge detection was performed using the Canny Edge Detection algorithm. This process highlights important boundaries and structures within the image.

5. Noise Addition and Noise Removal

Random Gaussian noise was added to the images to simulate image degradation. A Gaussian filter was then applied to reduce the noise and improve image quality. This demonstrated how image preprocessing techniques can help improve the quality of medical images.

Medical Imaging Dataset Exploration

Dataset 1: Chest X-Ray Pneumonia Dataset

Imaging Type: Chest X-Ray

Number of Images: 5,856

Available Classes:

- Normal
- Pneumonia

Dataset Imbalance:
The dataset contains more pneumonia images than normal images, which can create a class imbalance problem during model training.

Challenges Observed:

- Class imbalance
- Variations in image quality
- Differences in patient positioning

Dataset 2: Brain MRI Tumor Dataset

Imaging Type: MRI (Magnetic Resonance Imaging)

Number of Images: 253

Available Classes:

- Tumor
- No Tumor

Dataset Imbalance:
The dataset has slightly more tumor images than non-tumor images, although the imbalance is not very large.

Challenges Observed:

- Small dataset size
- Variations in tumor size and shape
- Differences in image resolution and quality